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BS730 - Spring 2026 Project 1

Project 1

The Optimizing Pain Treatment In Medical settings Using Mindfulness (OPTIMUM) clinical trial, completed in 2025, was a Pragmatic Clinical Trial (PCT) to inform clinicians, patients, administrators and policy makers how mindfulness can work in a real-life clinical setting, impact outcomes, increase access to non-opioid treatments and be reimbursable. It was conducted at three sites (Boston Medical Center, MA, UPMC, Pittsburgh, PA, and Piedmont Health Services, NC).

The trial involved an evidence-based 8-week medical group mindfulness clinical pain program (OPTIMUM) modeled on Mindfulness Based Stress Reduction (MBSR) and delivered in primary care settings through a telehealth and videoconferencing format.

A sample of 450 patients with chronic low back pain (cLBP) were randomized to either OPTIMUM or to primary care provider (PCP) usual care. This project will use the baseline data from patients at Boston Medical Center. The aims of the project are:

1. To determine the prevalence of high levels of pain
2. To examine the association between specific demographic factors and high levels of pain

A description of the variables in the dataset `optimum_base.csv` can be found on the next page. Missing data is denoted by NA.

References:

1. Greco CM, Gaylord SA, Faurot K, et al. The design and methods of the OPTIMUM study: A multisite pragmatic randomized clinical trial of a telehealth group mindfulness program for persons with chronic low back pain. *Contemp Clin Trials*. 2021;109:106545.
2. <https://optimum.pitt.edu/index.php/about-the-optimum-trial/>

Variable Descriptions

Variable	Description	Values
RECORD_ID	Record ID	Text String
PEG	A scale based on pain intensity and interference	Numeric, range 0-10
SEX	Sex at birth	1: Male, 2: Female, 3: Unknown, 4: Intersex
EDLEVEL	Highest level of education completed	1: Did not complete Secondary School or Less than High School; 2: Some Secondary School or High School Education; 3: High School or Secondary School Degree Complete; 4: Associate's or Technical Degree Complete; 5: College or Baccalaureate Degree Complete; 6: Doctoral or Postgraduate Education
EMPLOYMEN TSTAT	Current employment status	1: Full-time, 2: Not employed, 3: Part-time
PAINDUR_Y	Pain Duration in Years	Numeric
UNEMPL	Off work or unemployed for 1 month or more due to low-back pain	1: Agree, 2: Disagree, 3: Does not apply
RACEETH	Race/Ethnicity	1: Non-Hispanic White, 2: Non-Hispanic Black, 3: Hispanic, 4: Non-Hispanic Others, 5: Unknown ethnicity White, 6: Unknown ethnicity Black/African American, 7: Unknown ethnicity Others, 9: Unknown or not reported
AGEVAL	Participant age in years	Numeric
GENIDENT	Gender identity	1: Male, 2: Female, 3: Unknown, 4: Other, 5: Intersex
PROMIS29_G LOBAL07	Average pain rating	0-10 scale (0: No Pain, 10: Worst imaginable pain)
MED_OPIOID_ YN	At least one opioid reported at timepoint	0: No, 1: Yes
STATE	State	Text String
OVERALL_SVI _QUARTILE	Overall Social Vulnerability Index Quartiles	1, 2, 3, or 4 (Higher quartiles indicate greater social vulnerability)

Part 1. Data Import and Manipulation (25 points)

1. Read the dataset `optimum_base.csv` into R and assign it to a data frame called `optimum_base`. (5 points)

```
optimum_base = read.csv("optimum_base.csv", header=TRUE)
```

2. Create new variables in the `optimum_base` data frame and filter the data as follows:

a. Using the variable `PAINDUR_Y`, create a new variable called `PAIN_CAT` that takes on the values (5 points):

- 1: Less than 5 years of back pain ($\text{PAINDUR_Y} < 5$)
- 2: 5 to 15 years of back pain ($5 \leq \text{PAINDUR_Y} \leq 15$)
- 3: More than 15 years of back pain ($\text{PAINDUR_Y} > 15$)

```
optimum_base$PAIN_CAT = ifelse(optimum_base$PAINDUR_Y < 5, 1,  
                               ifelse(optimum_base$PAINDUR_Y <= 15, 2, 3))
```

b. A PEG score of 6 or more is considered moderate to severe pain. Create a variable called `PEG_HIGH` that takes on the value 0 if the participant has a baseline PEG score less than 6 ($\text{PEG} < 6$) and takes on the value of 1 if the PEG score is 6 or higher ($\text{PEG} \geq 6$). (5 points)

```
optimum_base$PEG_HIGH = ifelse(optimum_base$PEG < 6, 0, 1)
```

c. The variable `EDLEVEL` takes on 6 different values. Create a new variable called `COLLEGE_ANY` that takes on the value 0 if the participant has not had any college (`EDLEVEL` categories 1, 2 and 3) and the value 1 if the participant has had at least some college (`EDLEVEL` categories 4, 5 and 6). Note: `EDLEVEL` has missing value; assign NA to `COLLEGE_ANY` when `EDLEVEL` is missing. (5 points)

```
optimum_base$COLLEGE_ANY = ifelse(is.na(optimum_base$EDLEVEL), NA,  
                                  ifelse(optimum_base$EDLEVEL <= 3, 0, 1))
```

d. We will only be examining patients from Boston Medical Center. Filter the dataset to only keep patients from Massachusetts (`STATE` is equal to "MA"). (5 points)

```
optimum_base = subset(optimum_base, STATE == "MA")
```

Part 2. Creating Factor Variables with Labels (19 points)

1. In R, we use factors to create labeled categorical variables. Convert the following variables to factors with appropriate labels:

a. Convert PAIN_CAT to a factor with the following labels (5 points):

- 1 = "Less than 5 years of back pain"
- 2 = "Between 5 and 15 years of back pain"
- 3 = "More than 15 years of back pain"

Note: The labels should match the categories created in Part 1, Question 2a.

```
optimum_base$PAIN_CAT <- factor(optimum_base$PAIN_CAT, levels = c(1, 2, 3),  
labels = c("Less than 5 years of back pain", "Between 5 and 15 years of back  
pain", "More than 15 years of back pain"))
```

b. Convert EMPLOYMENTSTAT to a factor with the following labels (5 points):

- 1 = "Full-time"
- 2 = "Not employed"
- 3 = "Part-time"

```
optimum_base$EMPLOYMENTSTAT <- factor(optimum_base$EMPLOYMENTSTAT, levels =  
c(1, 2, 3), labels = c("Full-time", "Not employed", "Part-time"))
```

c. Convert PEG_HIGH, COLLEGE_ANY, and MED_OPIOID_YN to factors with labels "No" for 0 and "Yes" for 1 (5 points):

```
optimum_base$PEG_HIGH = factor(optimum_base$PEG_HIGH, levels = c(0, 1),  
                                labels = c("No", "Yes"))
```

```
optimum_base$COLLEGE_ANY = factor(optimum_base$COLLEGE_ANY, levels = c(0, 1),  
                                   labels = c("No", "Yes"))
```

```
optimum_base$MED_OPIOID_YN = factor(optimum_base$MED_OPIOID_YN, levels = c(0,  
1),  
                                    labels = c("No", "Yes"))
```

d. Convert OVERALL_SVI_QUARTILE to a factor variable with labels "1", "2", "3", "4". (4 points)

```
optimum_base$OVERALL_SVI_QUARTILE <-  
factor(optimum_base$OVERALL_SVI_QUARTILE, levels = c(1, 2, 3, 4), labels =  
c("1", "2", "3", "4"))
```

Part 3. Verifying the Analysis Dataset (6 points)

1. Use R code to display only the factor variables (PAIN_CAT, EMPLOYMENTSTAT, PEG_HIGH, COLLEGE_ANY, MED_OPIOID_YN, OVERALL_SVI_QUARTILE) and confirm they are correctly coded as factors. (6 points)

Hint: You can use indexing to select specific columns, e.g., `str(optimum_base[, c("VAR1", "VAR2")])`

```
str(optimum_base[, c("PAIN_CAT", "EMPLOYMENTSTAT", "PEG_HIGH", "COLLEGE_ANY",  
"MED_OPIOID_YN", "OVERALL_SVI_QUARTILE" )])
```

Part 4. Statistical Analysis (50 points)

Formatting Guidelines: Report percentages, means/SDs, and test statistics to 2 decimal places, and p-values to 4 decimal places (or "<0.0001" if smaller).

1. Complete Table 1 with summary statistics and p-values. (20 points)

- For **numerical variables** (e.g., age): Report mean (SD) for each group. Use the two-sample t-test with equal variance, for this question.
- For **categorical variables**: Report n (row %) for each level. Use the chi-square test without continuity correction (`correct = FALSE`). If any expected cell count is less than 5, use Fisher's exact test instead.

Note: Fisher's exact test is required for SVI quartile because >20% of expected cell counts are less than 5.

```
PEG_table = table(optimum_base$PEG_HIGH)
```

```
PEG_table
```

```
round(tapply(optimum_base$AGEVAL, optimum_base$PEG_HIGH, mean, na.rm =  
TRUE),2)
```

```
round(tapply(optimum_base$AGEVAL, optimum_base$PEG_HIGH, sd, na.rm = TRUE),2)
```

```
t.test(AGEVAL ~ PEG_HIGH, data = optimum_base, var.equal = TRUE)
```

```
college = table(optimum_base$COLLEGE_ANY, optimum_base$PEG_HIGH)
```

```
college
```

```
round(prop.table(college, 1) * 100,2)
```

```
chisq.test(college, correct = FALSE)$expected
```

```
chisq.test(college, correct = FALSE)
```

```
pain = table(optimum_base$PAIN_CAT, optimum_base$PEG_HIGH)
```

```
pain
```

```
round(prop.table(pain, 1) * 100,2)
```

```
chisq.test(pain, correct = FALSE)$expected
```

```
chisq.test(pain, correct = FALSE)
```

```
employed = table(optimum_base$EMPLOYMENTSTAT, optimum_base$PEG_HIGH)
```

```
employed
```

```
round(prop.table(employed, 1) * 100,2)
```

```
chisq.test(employed, correct = FALSE)$expected
```

```
chisq.test(employed, correct = FALSE)
```

```
opiod = table(optimum_base$MED_OPIOID_YN, optimum_base$PEG_HIGH)
```

```
opiod
```

```
round(prop.table(opiod, 1) * 100,2)
```

```
chisq.test(opiod, correct = FALSE)$expected
```

```
chisq.test(opiod, correct = FALSE)
```

```
SVI = table(optimum_base$OVERALL_SVI_QUARTILE, optimum_base$PEG_HIGH)
```

```
SVI
```

```
round(prop.table(SVI, 1) * 100,2)
```

```
chisq.test(SVI, correct = FALSE)$expected
```

```
fisher.test(SVI)
```

Table 1. Associations with High Levels of Back Pain (PEG score >= 6)

	High Back Pain: Yes	High Back Pain: No	P-value
	Yes (N=107)	No (N=44)	
Age (years) Mean (SD)	51.68 (12.80)	46.09 (14.43)	0.0201
At least some college n (%)			0.0873
Yes	33 (62.26)	20 (37.74)	
No	74 (75.51)	24 (24.49)	
Pain duration category n (%)			0.4670

	High Back Pain: Yes	High Back Pain: No	P-value
Less than 5 years	30 (69.77)	13 (30.23)	
Between 5 and 15 years	56 (68.29)	26 (31.71)	
More than 15 years	21 (80.77)	5 (19.23)	
Employment status n (%)			0.0799
Full Time	25 (59.52)	17 (40.48)	
Part Time	12 (63.16)	7 (36.84)	
Not Employed	69 (77.53)	20 (22.47)	
Opioid use n (%)			0.1381
Yes	19 (82.61)	4 (17.39)	
No	73 (66.97)	36 (33.03)	
Social vulnerability index quartile n (%)			0.0377
1	8 (47.06)	9 (52.94)	
2	8 (53.33)	7 (46.67)	
3	23 (79.31)	6 (20.69)	
4	67 (75.28)	22 (24.72)	

2. What proportion of the total sample has high level of back pain (PEG_HIGH = 1)? Is there an association between age and high level of back pain? First, use the F-test to check the equal variance assumption, then conduct the appropriate two-sample t-test. Report the null and alternative hypotheses, test statistic, degrees of freedom, p-value, and the mean (SD) age for each group as your measure of effect. State your conclusion. (15 points)

```
round(prop.table(PEG_table)*100,2)
```

```
var.test(AGEVAL ~ PEG_HIGH, data = optimum_base)
```

```
t.test(AGEVAL ~ PEG_HIGH, data = optimum_base, var.equal = TRUE)
```

Your answer here:

70.86% of the total sample has high level of back pain.

H0: The mean age is the same for high and low back pain.

H1: The mean age is not the same for high and low back pain.

$F = 1.27$, $p\text{-value} = 0.3245$

Since $p\text{-value} > 0.05$, we can assume equal variance.

$t = -2.35$

$df = 149$

$p\text{-value} = 0.0201$

The sample mean age for high back pain was 51.68 years ($n = 107$, $SD = 12.80$) compared to a mean of 46.09 years ($n = 44$, $SD = 14.43$) for low back pain.

Conclusion:

Since the $p\text{-value} < 0.05$, we reject the null hypothesis that the mean age is the same for the two groups.

We conclude that the average age of the population is statistically higher for high back pain compared to low back pain.

3. Is there an association between the social vulnerability index quartile (OVERALL_SVI_QUARTILE) and high levels of back pain (PEG_HIGH)? Report the null and alternative hypotheses, test statistic and degrees of freedom (if appropriate), p-value, and the proportion (%) with high back pain within each SVI quartile as your measure of effect. State your conclusion. Hint: Check if chi-square assumptions are met by examining expected cell counts using `chisq.test(table, correct = FALSE)$expected`. (15 points)

```
round(prop.table(SVI, 1) * 100, 2)
chisq.test(SVI, correct = FALSE)$expected
fisher.test(SVI)
```

Your answer here:

H0: There is no association between SVI quartile and high back pain.

H1: There is an association between SVI quartile and high back pain.


```
optimum_base$MED_OPIOID_YN = factor(optimum_base$MED_OPIOID_YN, levels = c(0, 1),  
                                     labels = c("No", "Yes"))
```

Part 1 question 2d

```
optimum_base$OVERALL_SVI_QUARTILE <-  
factor(optimum_base$OVERALL_SVI_QUARTILE, levels = c(1, 2, 3, 4), labels =  
c("1", "2", "3", "4"))
```

#Part 3 question 1

```
str(optimum_base[, c("PAIN_CAT", "EMPLOYMENTSTAT", "PEG_HIGH", "COLLEGE_ANY",  
"MED_OPIOID_YN", "OVERALL_SVI_QUARTILE" )])
```

#Part 4 question 1

```
PEG_table = table(optimum_base$PEG_HIGH)
```

```
PEG_table
```

```
round(tapply(optimum_base$AGEVAL, optimum_base$PEG_HIGH, mean, na.rm =  
TRUE),2)
```

```
round(tapply(optimum_base$AGEVAL, optimum_base$PEG_HIGH, sd, na.rm = TRUE),2)
```

```
t.test(AGEVAL ~ PEG_HIGH, data = optimum_base, var.equal = TRUE)
```

```
college = table(optimum_base$COLLEGE_ANY, optimum_base$PEG_HIGH)
```

```
college
```

```
round(prop.table(college, 1) * 100,2)
```

```
chisq.test(college, correct = FALSE)$expected
```

```
chisq.test(college, correct = FALSE)
```

```
pain = table(optimum_base$PAIN_CAT, optimum_base$PEG_HIGH)
pain
round(prop.table(pain, 1) * 100, 2)
chisq.test(pain, correct = FALSE)$expected
chisq.test(pain, correct = FALSE)
```

```
employed = table(optimum_base$EMPLOYMENTSTAT, optimum_base$PEG_HIGH)
employed
round(prop.table(employed, 1) * 100, 2)
chisq.test(employed, correct = FALSE)$expected
chisq.test(employed, correct = FALSE)
```

```
opiod = table(optimum_base$MED_OPIOID_YN, optimum_base$PEG_HIGH)
opiod
round(prop.table(opiod, 1) * 100, 2)
chisq.test(opiod, correct = FALSE)$expected
chisq.test(opiod, correct = FALSE)
```

```
SVI = table(optimum_base$OVERALL_SVI_QUARTILE, optimum_base$PEG_HIGH)
SVI
round(prop.table(SVI, 1) * 100, 2)
chisq.test(SVI, correct = FALSE)$expected
fisher.test(SVI)
```

#Part 4 question 2

```
round(prop.table(PEG_table)*100, 2)
var.test(AGEVAL ~ PEG_HIGH, data = optimum_base)
t.test(AGEVAL ~ PEG_HIGH, data = optimum_base, var.equal = TRUE)
```

#Part 4 question 3

```
round(prop.table(SVI, 1) * 100, 2)
chisq.test(SVI, correct = FALSE)$expected
fisher.test(SVI)
```